Polar Bookshop's primary services will use Spring. You'll see how to package a Spring application, run it as a JAR file, containerize it with Cloud Native Buildpacks, run it with Docker, and finally deploy containers with Kubernetes. You'll also see how to compile Spring applications to native images using Spring Native and GraalVM, and use them in serverless architectures, taking advantage of their instant startup time, instant peak performance, reduced memory consumption, and reduced image size. Then you'll deploy them on a serverless platform built on top of Kubernetes with Knative.

I'll show you how to automate the build stage by setting up a deployment pipeline with GitHub Actions. The pipeline will build the application at every commit, run the tests, and package it ready for deployment. Such automation will support a continuous delivery culture to bring value to customers quickly and reliably. In the end, you'll also automate the deployment of Polar Bookshop to the production Kubernetes cluster using GitOps practices and Argo CD.

UI

This book is focused on backend technologies, so I won't teach you any frontend subjects. Of course, your application will need a frontend for the users to interact with. In the case of Polar Bookshop, you'll rely on a client application using the Angular framework. I won't show you the UI application code in this book because it's out of scope, but I've included it in the code repository accompanying the book.

Summary